

YUHONG ZHAI

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SUMMARY

AI Engineer building production agent infrastructure at Meta — MCP plugins, tool-calling agents, multi-turn workflows, eval harnesses, and agent observability. 6 years of production engineering across Meta, Walmart, and Nike serving billions of users.

WORK EXPERIENCE

AI Engineer / Data Engineer, Meta Platforms Inc. Menlo Park, CA *Aug. 2024 - Present*
Monetization Payments Org (WhatsApp & Messenger Payments — LATAM & India)

- Led design and shipped an internal Claude-based MCP plugin as first owner, exposing modular agent skills via tool-calling interfaces (dashboard migration, monitoring, governance); adopted by 10 teams and 200+ engineers, replacing manual analytics workflows org-wide.
- Built Metric Portfolio, an event-driven agent observability framework with custom eval harnesses that uses LLM prompts to analyze code diffs, dashboards, and upstream loggers in real time — catches ~2 major SEVs per month pre-production, preventing multi-week metric outages.
- Developed automated health-check agents with multi-step tool calling and structured outputs to classify failure root causes — reduced manual triage from weeks to hours across global payments teams.
- Designed prompt-based anomaly detection with tiered alerting and asymmetric thresholds, integrated with Claude Infra and LLM agents to monitor 50+ business metrics — reduced time-to-detection 15%, cut false positives 50%.
- Architected the first user-engagement funnel analytics for the org across 36 entry points, \$33B TPV, and 263M+ monetizable transactions — reviewed daily by the director and adopted as the canonical funnel tool across product and engineering teams.
- Shipped end-to-end data foundation and semantic data layer for business messaging payments in 10 weeks — distributed SQL pipelines serving billions of users, 24 new monetization metrics to executive dashboards.

Reality Labs — Wearable Devices Team

- Built unified connectivity and power-domain data pipelines as a horizontal workstream — reduced p95 telemetry latency from 15 min to 90 sec, cut compute spend ~40%, and initiated a data quality framework for proactive regression detection.
- Onboarded and mentored 2 new-hire engineers on pipeline architecture, data quality, and on-call ownership.

Data Engineer, Walmart Connect, San Bruno, CA *Oct. 2021 - Aug. 2024*

- Built streaming ETL pipelines processing 2B+ events for Walmart Connect advertising analytics, powering real-time advertiser dashboards across the Walmart Marketplace ads platform.
- Optimized SQL queries and indexing strategy for analytics dashboards, reducing query latency from minutes to seconds.
- Analyzed OLQ score impact on Gross Merchandise Value across 14k Walmart Marketplace sellers over 18 months; identified the top 10% contributing 90% of GMV.

Data Engineer, Nike, Beaverton, OR *Aug. 2020 - Oct. 2021*

- Built metric-driven dashboard and alerting for the Audience Workbench platform using Splunk and ServiceNow, providing visibility into 70 audiences/day and 500/week for external stakeholders.
- Ingested and validated ~1B consumer behavior events/day into Nike's data ecosystem via complex Snowflake SQL queries powering NikeApp, NIKE.COM, and SNKRS.
- Developed Python and SQL automation in Databricks for data validation, anomaly detection, and exploratory analysis — reduced recurring analytical workflows from weeks to hours.

SKILLS

Languages: Python, SQL

AI & LLM Systems: MCP (Model Context Protocol), LLM APIs (Claude, OpenAI), prompt engineering, tool-calling agents, AI agent orchestration, multi-turn reasoning, RAG, eval harnesses, agent observability, structured outputs

Data & Infrastructure: distributed SQL pipelines, Snowflake, Databricks, Airflow, Docker, AWS, GCP, Git

EDUCATION

Texas A&M University, College Station, TX — M.S. in Quantitative Economics and Econometrics *May 2019*

Southwestern University of Finance and Economics, Chengdu, China — B.S. in Economics *June 2017*